Inverse relations and functions

1 $y = x$

2

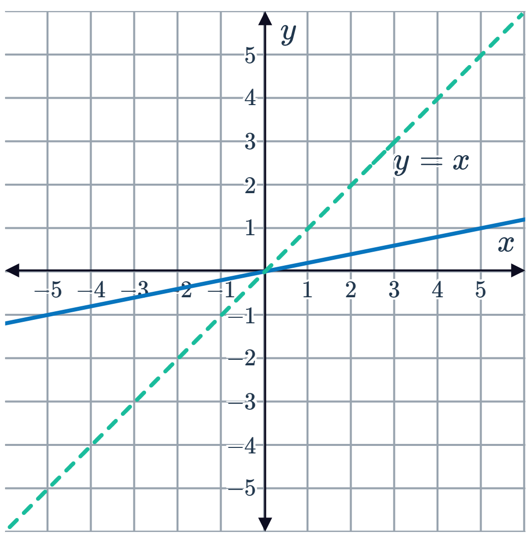
a No	**b** Yes	**c** No	**d** Yes
e No	**f** Yes	**g** No	**h** Yes

3

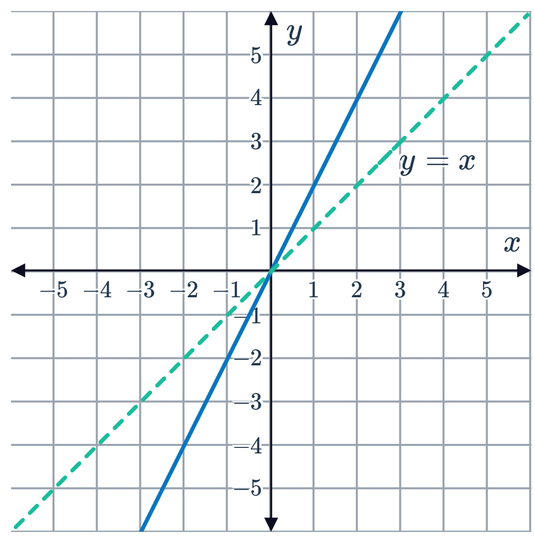
a Yes	**b** No	**c** Yes	**d** Yes
e No	**f** No	**g** Yes	**h** Yes
i Yes	**j** Yes	**k** No	

4

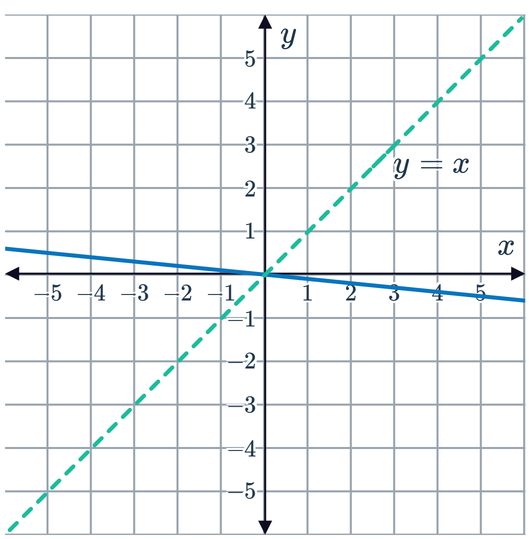
a



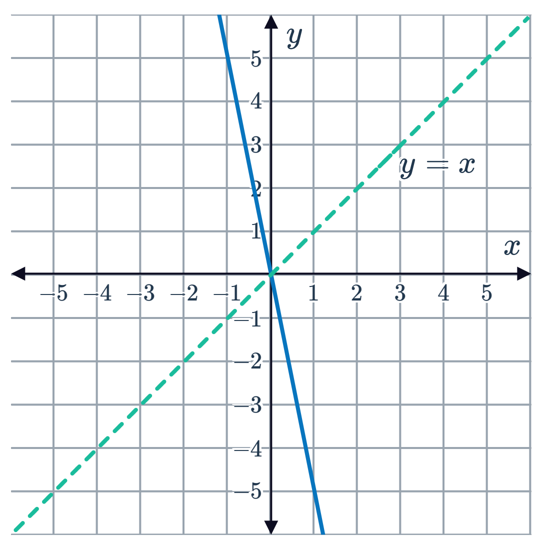
b



c



d



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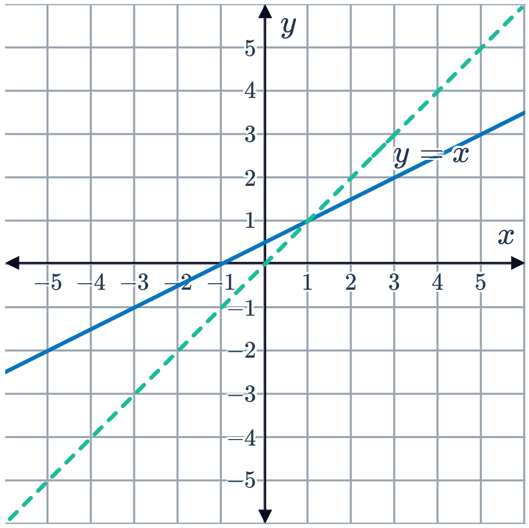
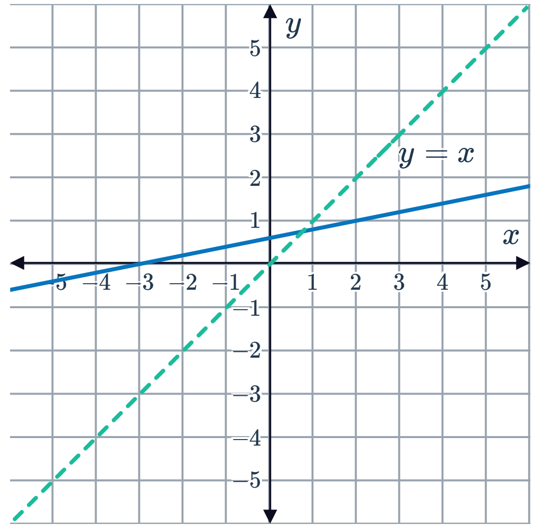
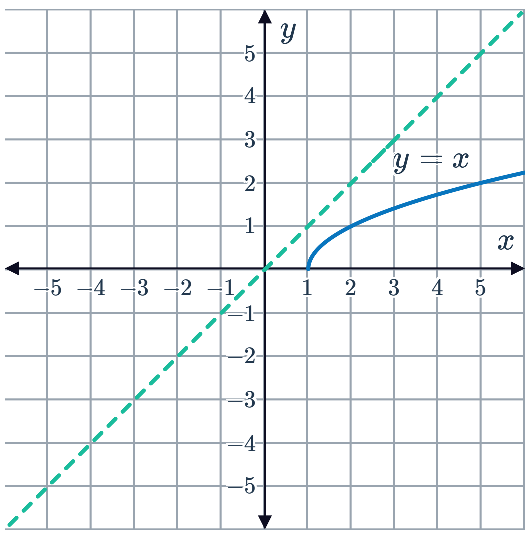
e

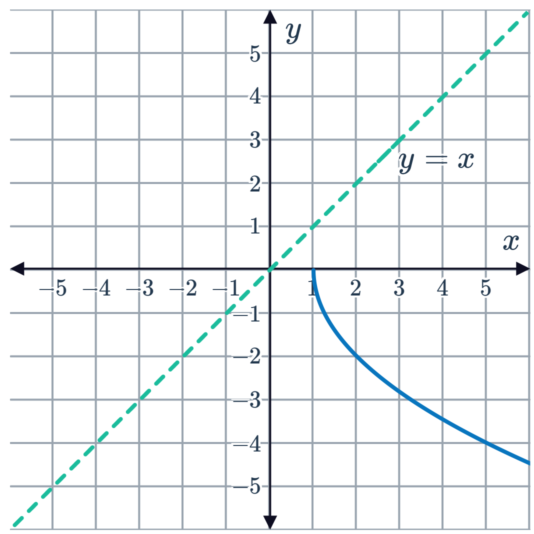




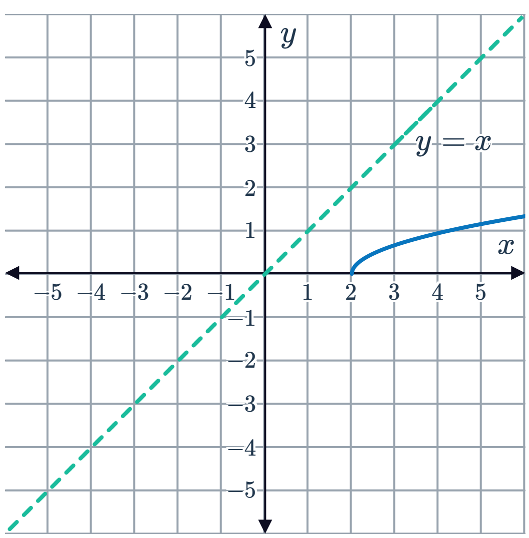
g



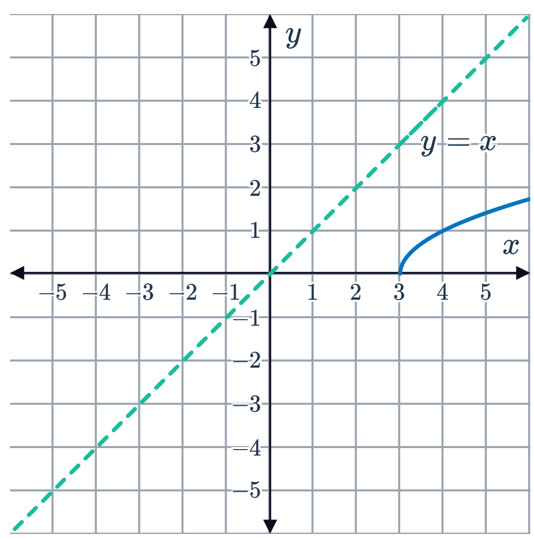
h



i



j



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5

a Function

b Relation

6

a

i x

ii x

iii Yes

b

i $x-12$

ii $x-4$

iii No

c

i x

ii x

iii Yes

7

a $y=\frac{x}{5}$

b $y=\frac{x+8}{7}$

c $y=\frac{3}{4}(x+5)$

d $y=\sqrt[3]{\frac{x}{6}}$

e $y=8(1-x)$

f $y=6x+7$

g $y=x^3+2$

h $y=\sqrt[3]{x-3}+4$

i $y=-\frac{2x}{x-5}$

j $y=\frac{8}{x-5}+9$

k $y=\frac{8}{5x}-\frac{3}{5}$

l $y=\sqrt[3]{\frac{7}{x+3}}+8$

m $y=\frac{\sqrt[3]{\frac{x+7}{4}}+2}{9}$

Domain and range

8

a $\{-3\}$

b $\{6\}$

c $(6,-3)$

d $\{6\}$

e $\{-3\}$

f The domain of f is equal to the range of f^{-1}, and the range of f is equal to the domain of f^{-1}.

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9

a $\{6,-5\}$

b $\{6,-5\}$

10

a $\{5,7,8\}$

b $\{9,-4,2\}$

c $\{(9,5),(-4,7),(2,8)\}$

d $\{9,-4,2\}$

e $\{5,7,8\}$

11

a $a=-\dfrac{1}{2}, b=\dfrac{5}{2}$

b $[5,0]$

c $[-5,5]$

12

a

i $f^{-1}(x)=\dfrac{1}{5}(x-15)$

ii $(-\infty,\infty)$

iii $(-\infty,\infty)$

b

i $f^{-1}(x)=\dfrac{1}{3}(4-x)$

ii $(-\infty,\infty)$

iii $(-\infty,\infty)$

c

i $f^{-1}(x)=\dfrac{1}{8}(x+9)$

ii $[-41,7]$

iii $[-4,2]$

d

i $f^{-1}(x)=\sqrt{x}$

ii $[0,\infty)$

iii $[0,\infty)$

e

i $f^{-1}(x)=\sqrt{16-x^2}$

ii $[0,4]$

iii $[0,4]$

13

a

i $f(x)=\dfrac{2x}{3}-2$

ii $g(x)=\dfrac{3x}{2}+3$

b

i x

ii x

c

i True

ii True

iii False

iv True

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14

a $[-4, 9]$ b $[-4, 5]$ c $[-4, 5]$ d $[-4, 9]$

e False

15

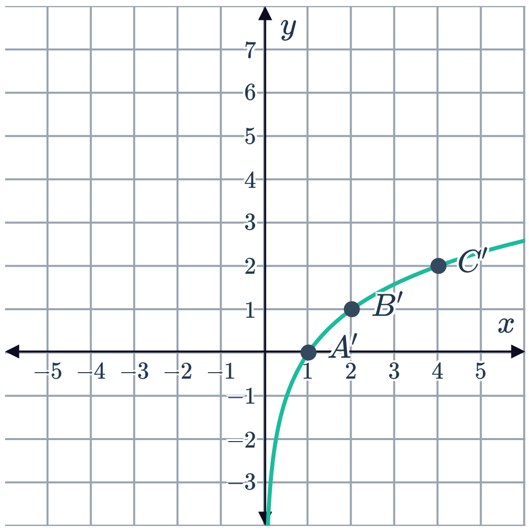
a

i $(0, 1)$ ii $(1, 2)$ iii $(2, 4)$

b

i $(1, 0)$ ii $(2, 1)$ iii $(4, 2)$

c



16

a $(-\infty, \infty)$

b $(0, \infty)$

c $(0, \infty)$

d $(-\infty, \infty)$

e The range of f^{-1}

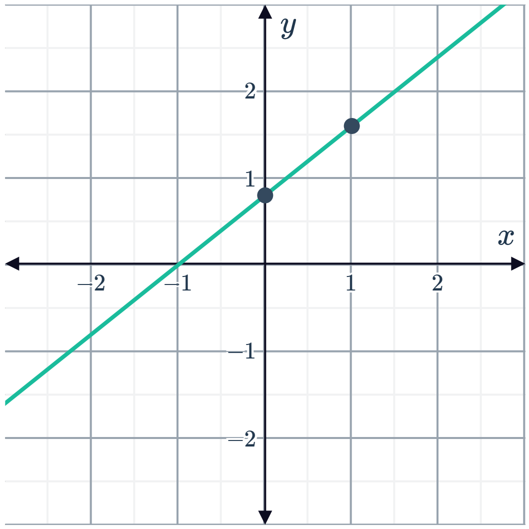
f The domain of f^{-1}

17

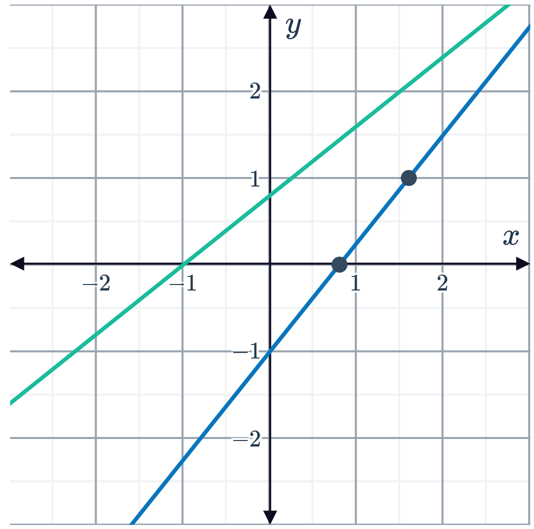
a $x \geq 2$ b $y \geq -5$ c $y = x^2 - 4x + 9$

18

a



b



c $(-\infty, \infty)$

d $(-\infty, \infty)$

e $(-\infty, \infty)$

f $(-\infty, \infty)$

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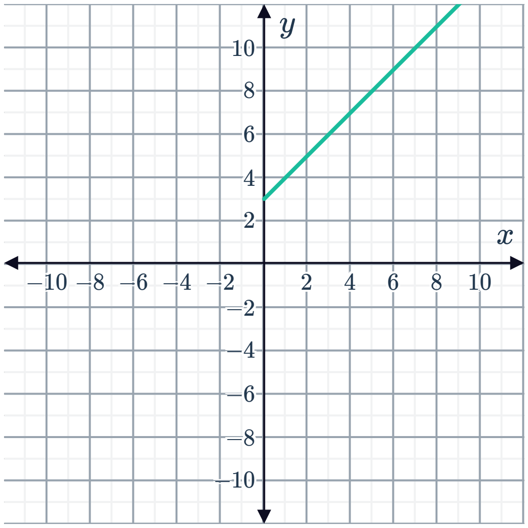
19

a

i

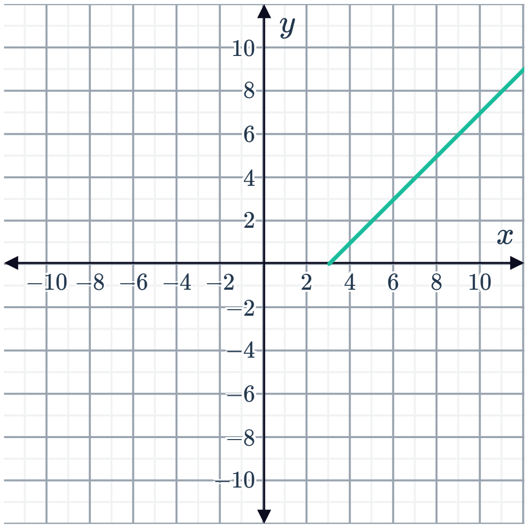


ii $f^{-1}(x) = x - 3$

iii $[3, \infty)$

iv $[0, \infty)$

v



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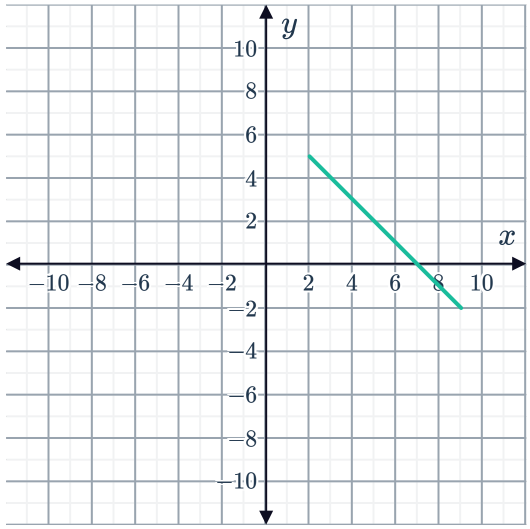
b

i

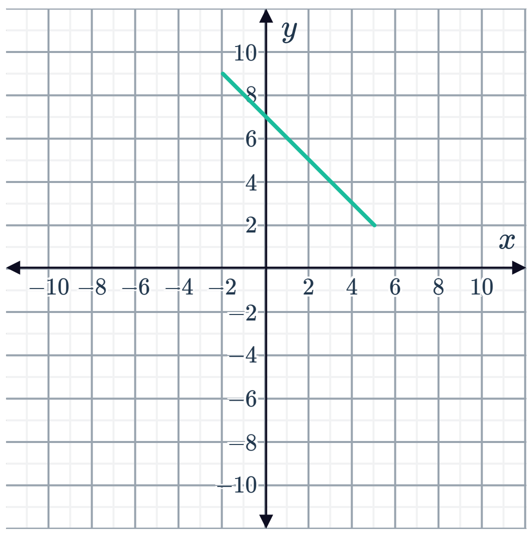


ii $f^{-1}(x) = 7 - x$

iii $[-2, 5]$

iv $[2, 9]$

v



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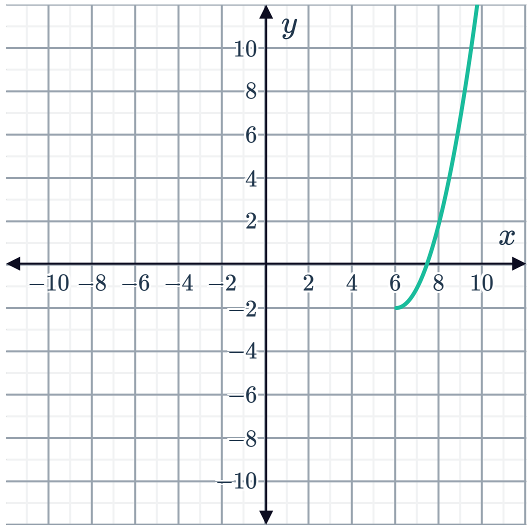
c

i

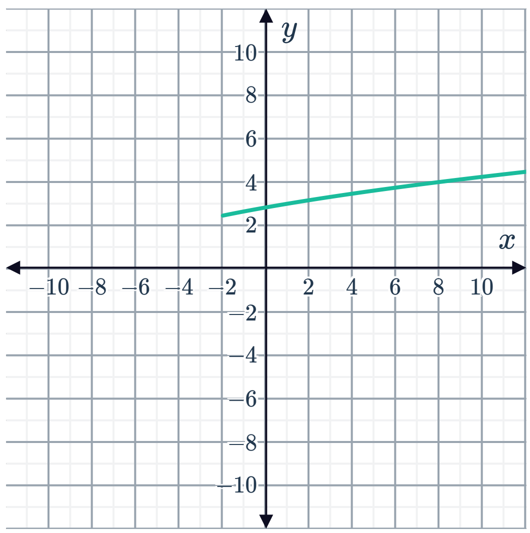


ii $f^{-1}(x) = \sqrt{x+2} + 6$

iii $[-2, \infty)$

iv $[6, \infty)$

v



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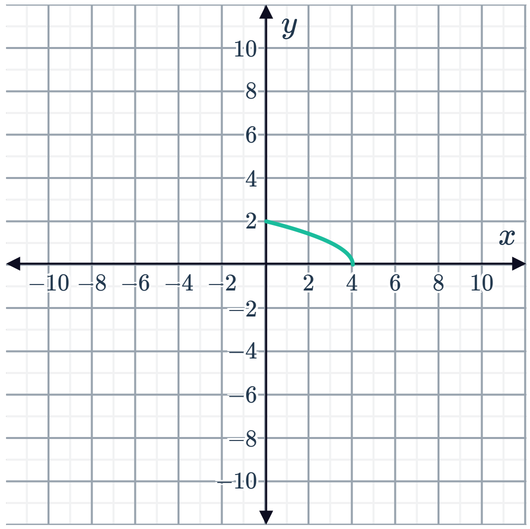
d

i

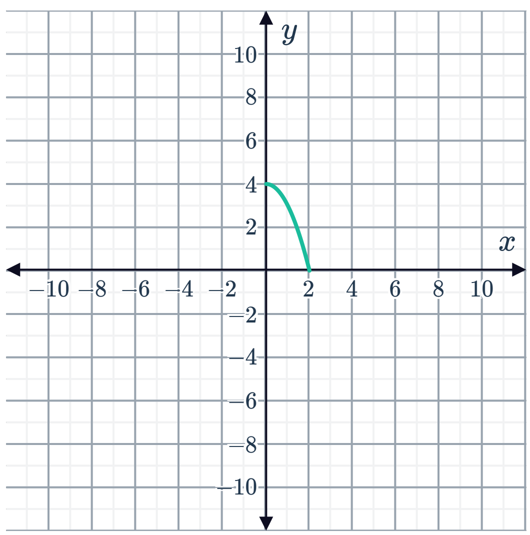


ii $f^{-1}(x) = 4 - x^2$

iii $(0, 2]$

iv $[0, 4)$

v



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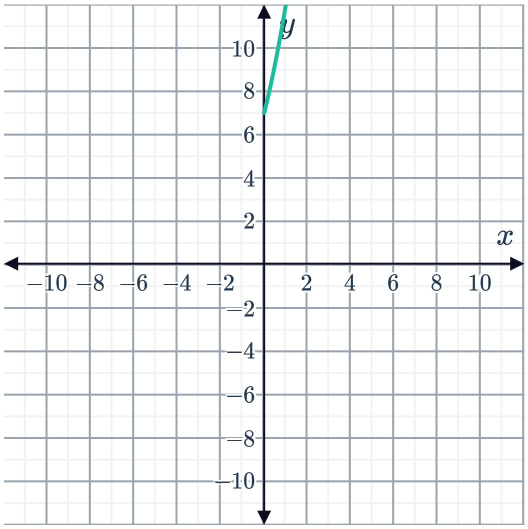
e

i

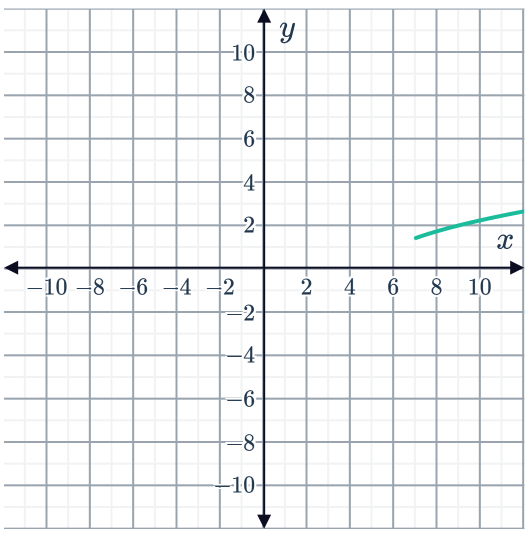


ii $f^{-1}(x) = \sqrt{x-3} - 2$

iii $[7, \infty)$

iv $[0, \infty)$

v



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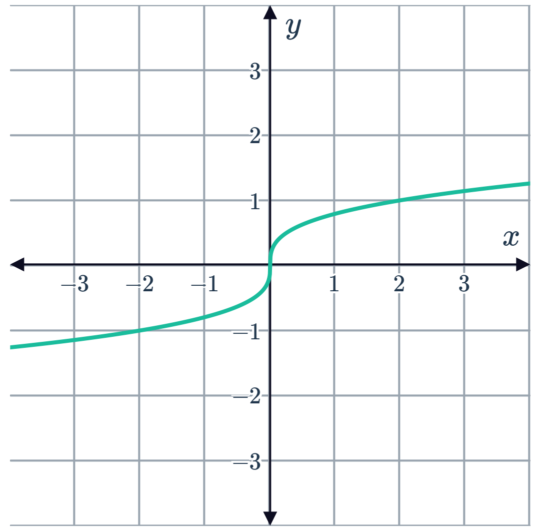
20

a Yes

b All real x

c All real y

d



21

a No

b All real x

c $x > 2$ and $x < -2$

d $y > 2$ or $y < -2$

22

a $T = 1.07A$

b $A = \dfrac{T}{1.07}$

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